Security and Privacy for Machine Learning (Honors)

Internal Assessment 1

Problem Formulation and Preliminary Implementation

Backdooring the Guard

A Data-Poisoning Attack on a CAN-Bus Intrusion Detector,
and a Layered Privacy Defence (DP-SGD, Local DP, Compressive Privacy)

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An individual integrated research work spanning all five course modules.

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Rubric-compliance map. The report mirrors the five IA-1 criteria.

IA-1 criterion	Evidence	Section
Problem & module mapping	Problem, objectives, scope, five-module mapping	§1
Conceptual understanding	Concepts, workflow, implement/analyse split	§2
Dataset preparation	HCRL/OTIDS source, windowing, provenance	§3
Initial implementation	Honest detector + first backdoor demonstration	§4
Preliminary evaluation	Observations, evaluation plan, contribution	§5

1 Problem Definition and Module Mapping

1.1 Problem statement

An in-vehicle Controller Area Network (CAN) carries safety-critical messages with *no authentication* (Koscher et al., 2010): any node, or an attacker at the OBD-II port, can inject frames. The standard defence is a machine-learning *intrusion detection system* (IDS) that flags injected traffic. But the IDS is itself a model trained on logged CAN data, so its training pipeline is an attack surface. This project asks: *can an attacker poison the IDS's training so that, whenever a secret trigger pattern is present, the detector silently passes real attacks as "Normal"* (Gu et al., 2017) – and which privacy mechanisms from the course both defend against this and protect the telemetry.

1.2 Significance, objectives and scope

A backdoored safety monitor is a worst-case failure: it looks perfect in testing yet goes blind on command. **Objectives:** (O1) build a CAN-IDS and a trigger-based backdoor; (O2) study differentially private training as a defence; (O3) add local DP, output modification and compressive privacy to cover the collection, release and feature-sharing exposures; (O4) measure the privacy–utility trade-off throughout. **Scope:** a binary normal-vs-attack IDS on a real HCRL CAN dataset, attacked and defended in a sandbox; no live vehicle is involved.

1.3 Mapping to all five modules

Table 1: Mapping of the problem to the five modules (all anchored to the CAN-IDS).

Module	Concepts used	Role in this project
M1 Privacy considerations	Threats/attacks on ML (1.2); membership inference (1.1)	Backdoor/poisoning attack; MIA on the IDS
M2 Differential privacy	DP-SGD supervised training (2.3); DP k-means (2.4)	Private IDS training; does DP blunt the backdoor
M3 Local DP	LDP mechanisms (3.1–3.3); LDP Naive Bayes (3.4)	Randomise CAN features at the source
M4 Privacy-preserving data mining	Modify output (4.5); k-anonymity (4.3)	Output masking vs MIA; CAN-ID k-anonymity (analytical)
M5 Compressive privacy	Mechanisms + utility (5.1–5.4)	PCA/DCA compress the window before sharing

2 Conceptual Understanding

Backdoor (data-poisoning) attack. Injecting a small number of training examples that carry a trigger and are mislabelled teaches the model a hidden rule "trigger ⇒ attacker's label" while leaving clean-input behaviour intact (Gu et al., 2017). Here the trigger is a rare CAN ID stamped into a few frames of a window – physically injectable because CAN is unauthenticated. **Differential privacy / DP-SGD.** DP-SGD (Abadi et al., 2016) clips each example's gradient and adds Gaussian noise, bounding any single example's influence. This both limits membership inference (Shokri et al., 2017) and directly attacks the mechanism a few poisoned windows rely on, so it is a natural candidate defence. **Local DP** (Duchi et al., 2013; Wang et al., 2019) randomises values at the source; **compressive privacy** (Kung, 2017) projects features to a low-dimensional code that keeps utility but discards reconstructable raw signal.

What is implemented vs analysed. The backdoor, the CNN/RF detectors, DP-SGD, the LDP mechanisms, output modification and PCA/DCA are *implemented*; classical table k-anonymity is treated *analytically* because a CAN message stream is not a static published table.

2.1 Workflow

Figure 1 shows the pipeline: CAN frames become windows, a 1-D CNN is trained as the honest IDS, poisoning implants the backdoor, and the privacy mechanisms are then layered on and measured.



Figure 1: Preliminary attack pipeline (privacy defences added in IA-2).

3 Dataset Identification and Preliminary Preparation

Source. HCRL in-vehicle CAN intrusion captures (OTIDS family) via the public Kaggle mirror `bikashkundu/can-hcrl-otids`: one normal capture and three attack captures. **Format:** `TS, ID1, ID0, LEN, DLC0-7, target`. **Suitability:** real, labelled vehicle-network traffic with a clear normal/attack split – exactly what a CAN-IDS needs, and the trigger can be a CAN ID absent from the data. **Preparation:** each frame $\rightarrow [\Delta t, \mathrm{ID}, \mathrm{DLC}, b_0 \dots b_7]$; windows of 100 frames (stride 50), labelled by capture class, with a time-contiguous split *inside* each capture to prevent window leakage; standardised on training statistics. Full provenance (row counts, SHA-256 hashes, licence) is in `data/PROVENANCE.md`.

4 Initial Implementation and Demonstration

Code is in `src/canids/` (mechanisms from scratch; Opacus only for the DP-SGD accountant); pipeline correctness is checked in `tests/` (9 passing tests). The complete codebase is at `https://github.com/aish475/MLCS_IA`.

4.1 Honest detector

Table 2 shows the baseline IDS detects CAN intrusions almost perfectly (CNN $F_1 = 0.996$). A near-perfect detector is the right starting point: the project's claim is that even this can be silently backdoored.

Table 2: Honest intrusion detectors (preliminary).

Detector	Acc.	Bal. acc.	Prec.	Recall	F_1
RandomForest	0.946	0.948	1.000	0.897	0.946
1D-CNN	0.996	0.996	0.992	1.000	0.996

4.2 First backdoor demonstration

Table 3 and Figure 2 give the first poison sweep: the honest model ignores the trigger (ASR ≈ 0.000), but once a small fraction of attack windows are poisoned, the trigger flips gen-

uine attacks to “Normal” at a high rate while clean accuracy stays near the honest baseline.

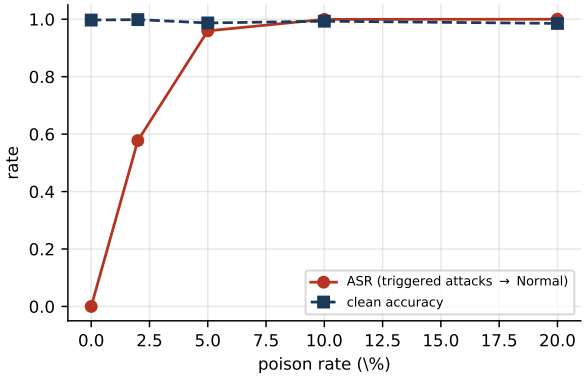


Figure 2: Clean accuracy and ASR vs poison rate.

Table 3: Backdoor poison sweep (preliminary).

Poison rate	Clean acc.	ASR
0%	0.997	0.000
2%	0.999	0.578
5%	0.987	0.959
10%	0.993	0.999
20%	0.985	1.000

5 Preliminary Evaluation and Documentation

5.1 Observations

- The honest CNN is an almost perfect detector, so any successful backdoor is, by construction, invisible on clean traffic.
- The trigger causally affects only the poisoned model – the honest model's response to the trigger is near zero – which isolates the attack from ordinary misclassification.

5.2 Evaluation plan for IA-2

Detection: accuracy, balanced accuracy, attack precision/recall/F_1. Backdoor: ASR, clean-accuracy gap, poison-rate sweep. Privacy: ε (at $\delta = 10^{-5}$), ASR-vs-ε and membership-inference AUC under DP-SGD; LDP estimator utility vs local budget; compressive F_1 vs dimension. All over multiple seeds.

5.3 Individual-contribution statement

I, **Aishwarya Nitin Gawali** (16014223007, B1), designed the backdoored-CAN-IDS study and wrote the `src/canids` package from scratch – the HCRL parser and time-contiguous windowing, the 1-D CNN (GroupNorm, DP-SGD-ready) and RandomForest detectors, the trigger-injection and poisoning code with the ASR measurement, and the scaffolding for the local-DP, compressive and membership-inference components completed in IA-2 – together with the unit tests and the experiment driver. Opacus is used only as the DP-SGD accountant.

References

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